Correct-field F4 checks and failed signature controls

E25; 2026-10-23; local bounded evidence, L0.

The pilot requests 48 curve/arity/encoding/order cells. Twenty-four grevlex cells execute five engines; 24 lex cells are unsupported by this signature backend and retain an independent affine reference.

Stock affine F4, the output-adapted affine F4, and homogeneous F4 all return parsed bases in 24 cells. Every dehomogenized affine basis passes a complete independent leading-ideal/inclusion certificate. Homogeneous completeness passes 21 cells; three remain checker-limited.

Both signature arms fail all 24 cells and the small positive control, with signal exits and no parsed bases. The source restores the actual field before its signature core; a placeholder field diagnostic is not evidence of wrong-field computation. The local patch changes result serialization only.

Earlier parser errors and manual S-pair timeouts remain recorded. The manual checker printed all 49 summary rows, but its shell receipt tail failed; no missing successful receipt is invented. The separate leading-ideal replay has a clean receipt.

Independent certificate	Accepted cells / 24
dehomogenized_affine_complete_basis_gate:A0	24
dehomogenized_affine_complete_basis_gate:H0	24
dehomogenized_affine_complete_basis_gate:S0	24
homogeneous_complete_basis_gate:H0	21

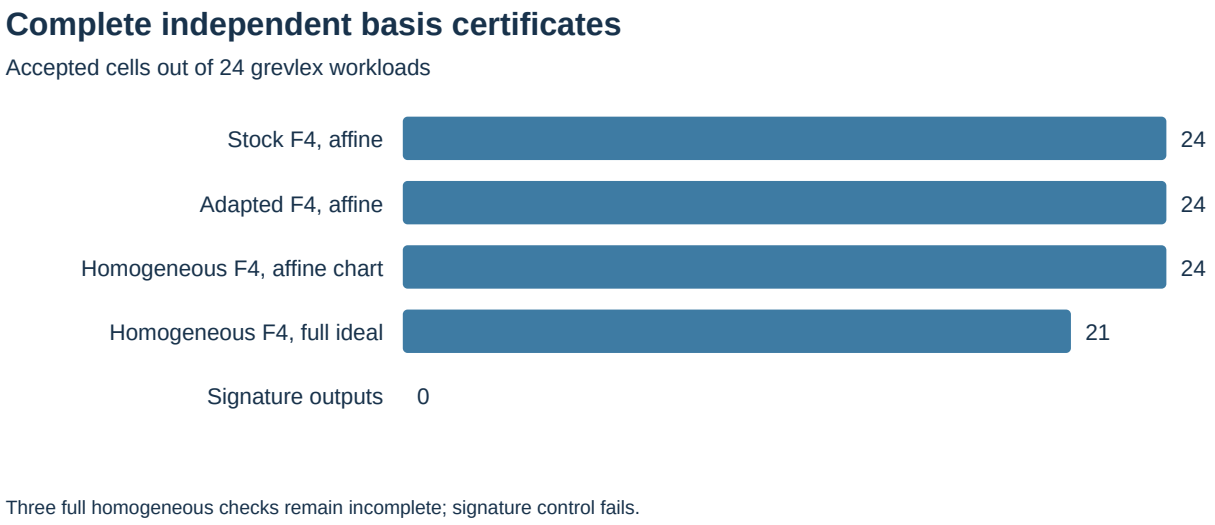

Figure 1: E25: completeness is checked, not inferred from ideal containment or solver exit status.

No timing or speed comparison is admitted. Magma/FGb/OpenF4 and an applicable successful signature reference remain missing. The historical gate key “dimension” records Singular vdim (quotient vector-space dimension), not Krull dimension.

Verified locally; complete raw publication awaits artifact-store access. No calibrated speedup, novelty, independent-host replay or NIST DLP is established.

Native assessment: ../../execution-v2/assessment-20261023-v3/ASSESSMENT.json. Raw members: runs/homogeneous-20261023-v1, -v2, -recheck-v3 and -leading-v4. Source/input/binary hashes and exact argv: E25-receipts.json and ../../execution-v2/receipts/.

All phase receipts, including errors and timeouts: E25-receipts.json. The complete execution ledger and original protocols retain actual limits and missing measurements.